

The Seven Basic Quality Tools: Getting Started with Data Visualizations

By

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A common first step in the learning journey of a quality engineer is mastering a set of data visualization tools called the “Seven Basic Quality Tools.” Popularized by an architect of the modern quality movement, Karou Ishikawa, the seven basic quality tools allow practitioners to readily extract useful insights from raw data and build a foundation for further statistical analysis. While simple to use, these tools stand as pillars of systematic problem-solving, aiding in the identification, analysis, and improvement of processes.

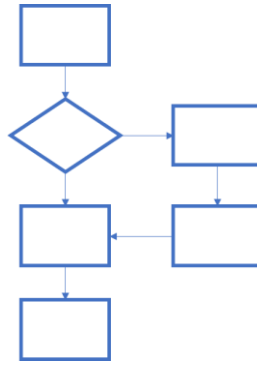
The seven basic quality tools, in order of complexity, are: Check Sheets, Flowcharts, Cause and Effect Diagrams, Histograms, Pareto Diagrams, Scatter Plots, and Control Charts.

Check sheets are simple, yet effective tools for collecting and organizing data. Technicians and inspectors use them to systematically record and tally information, facilitating the identification of patterns and trends. Whether monitoring machine downtime, defects, or any other measurable variable, check sheets offer a structured approach to data collection, providing a solid foundation for further analysis.



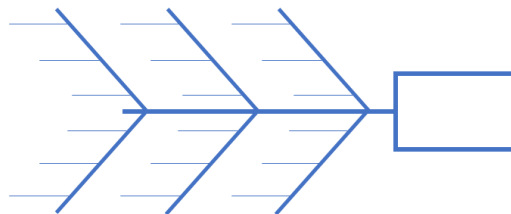
On a forging press, checklists can be used to highlight the steps needed to set up a tooling block or to verifying that preventive maintenance tasks have been completed.

Flowcharts provide a visual representation of the steps in a process, helping engineers identify inefficiencies, bottlenecks, and opportunities for improvement. By mapping out the sequence of activities, engineers can streamline processes, reduce waste, and enhance overall efficiency. Flowcharts serve as a valuable communication tool, facilitating a shared understanding of complex processes among team members. Flowcharts also set the stage for more elaborate process mapping tools like Swim Lane Diagrams and Value Stream Maps.



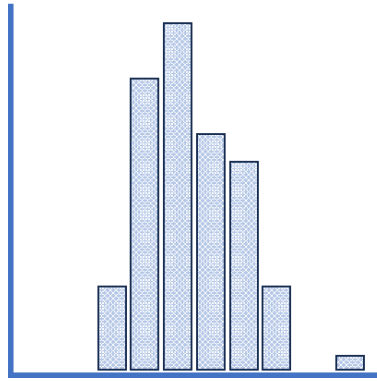
For organizations working within automotive, aerospace, heavy truck or similar supply chains that follow the Production Part Approval Process (PPAP), a type of flowchart called the process flow diagrams is used by suppliers to communicate to their customers a high-level view of their manufacturing process.

Cause-and-effect diagrams – also eponymously named Ishikawa Diagrams after distinguished management theorist, Kaoru Ishikawa -- are powerful tools for identifying and organizing potential causes of a problem. By visually mapping out the factors influencing a specific outcome, engineers can systematically analyze and address the root causes of issues. This tool promotes cross-functional collaboration and helps teams gain a comprehensive understanding of complex problems, fostering effective problem-solving. Cause-and-effect diagrams are also commonly used in conjunction with another problem-solving framework called the “Six M’s”, referring to the six major sources of process variation: Machine, Method, Manpower, Material, Measurement, and Mother Nature.



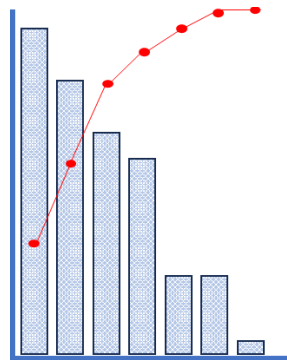
And of course, the original “Manpower” and “Mother Nature” could readily be replaced by less gender-specific terms such as “associate” and “ambient conditions.”

Histograms are graphical representations of data distributions, offering analysts visual insights into the variability of a process. By displaying data in a bar chart format, histograms reveal the frequency distribution of specific characteristics, enabling engineers to pinpoint areas for improvement. Engineers can use histograms to identify patterns, outliers, and potential root causes of issues affecting the quality of a product or process.



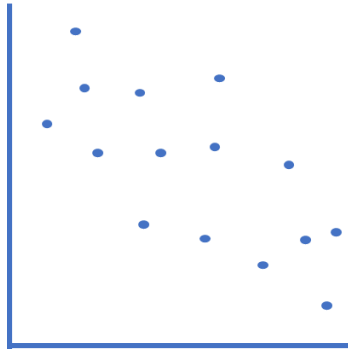
Within forging companies, histograms are commonly used to display the results of a dimensional study of a key product characteristic such as a flange thickness or overall length.

Expanding upon the so called “80/20 rule”, Pareto analysis and its associated visualization the Pareto diagram help problem-solvers identify and prioritize the most critical issues affecting a process. By focusing efforts on the vital few factors that contribute the most to a problem, engineers can maximize the impact of their improvements. This tool aids in decision-making, ensuring that resources are allocated efficiently to address the primary contributors to quality issues.



A common application of Pareto analysis in manufacturing quality is for the visualization of product defect data. For instance, a quality manager may summarize the quantity of each defect type such as underfill or conveyance damage found each month, and plot their findings in the form of a Pareto diagram.

Scatter diagrams are used to identify the relationship between two variables, helping engineers understand the correlation between factors that may impact product quality. By plotting data points on a graph, engineers can visually assess whether changes in one variable corresponds to changes in another. This tool aids in identifying potential cause-and-effect relationships and guides targeted interventions for process improvement.



Scatter plots may be used to depict a wide variety of relationships common in forging processes. For instance, in thermomechanical simulations designed to find the optimal forging temperature for a given alloy, test engineers plot a range of billet temperatures on the x-axis and the resulting compressive stresses on the y-axis. The temperature at which the compressive stress is minimized is often a good starting point for a forging process.

Control charts are instrumental in monitoring and maintaining the stability of a process over time. By plotting data points against control limits statistically generated from historical data, engineers can distinguish between common cause and special cause variation. This tool enables timely detection of deviations from the norm, allowing for prompt corrective actions to maintain consistent quality levels in processes.



Control charts also serve as a foundation for more advanced quality tools such as process capability analysis. Together, these tools are designed to compare the specification range of a given feature to its normal limits within a particular process.

In the intricate world of engineering, the Seven Basic Quality Tools underpin systematic problem-solving and process improvement. By leveraging these tools, engineers can analyze data, identify trends, and make informed decisions to enhance the quality of their products and processes. As technology advances and industries evolve, mastering these foundational tools remains essential for delivering excellence in every facet of quality.

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